Fiscal Year:	FY 2007	**Task Last Updated:**	FY 10/28/2007
PI Name:	Bearman, Gregory Ph.D.		
Project Title:	Advanced Trace Gas Monitoring Technology for NASA Human Space Flight		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Greg.Bearman@jpl.nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	818-354-3285
Organization Name:	NASA Jet Propulsion Laboratory		
PI Address 1:	Biomolecular Systems Research Program		
PI Address 2:			
PI Web Page:			
City:	Pasadena	**State:**	CA
Zip Code:	91109	**Congressional District:**	29
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Space Biology 02-OBPR-01
Start Date:	11/11/2002	**End Date:**	11/11/2006
No. of Post Docs:	4	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
	This project is a joint Rice University- George Washington University-JPL research and development program of mid-infrared diode laser sources in the 3 to 4.4 micron region and their application to trace gas sensing in NASA applications relevant to spacecraft environmental monitoring and advanced life support. Presently, single frequency diode and quantum cascade lasers are not available in the proposed mid infrared spectral region that covers an important range where sensitive, selective and fast trace gas detection can now only be performed with complex spectroscopic sources, such as lead salt diode lasers and indirect laser sources based on difference frequency generation. Pulsed and continuous wave Type I quantum-cascade lasers, now commercially available cover the wavelength region from 4.4 - 24 microns. However, there is a significant need to access wavelengths from 3 to 4.4 microns, where a number of critical molecules (e.g. CH4, HCl, NO2, H2CO,SO2, and CO2) posses strong ro-vibrational lines. The Laser Science group at Rice University has accumulated a decade of experience in laser based trace gas detection in applications directly		

Task Description:	relevant to NASA as well as in other areas such as atmospheric chemistry, environmental monitoring, industrial process control and medical diagnostics (http://www.ece.rice.edu/). The group of Prof. Houston Miller is providing expertise in spectroscopic strategy for detecting Code M molecular target gases as well as developing a cavity ringdown spectroscopy (CRDS) based sensor technology. JPL with world-class semiconductor facilities and the expertise necessary is developing Sb-based diode and type II interband cascade lasers with distributed feedback structures for meeting the application needs. At Rice the primary emphasis will be to demonstrate and emphasize a novel optical detection technique based on an ultracompact photoacoustic analyzer design that will significantly enhance the performance and robustness of near infrared and mid-infrared gas sensors. Early in the second year we will select one technology for further development. Our goal is to develop a prototype sensor device that can be packaged into a pelican case and can run off a laptop and USB connection. The laser will be a cw DFB that can be operated with a TEC.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	CRDS:During the last year, we have demonstrated the measurement of several air constituents at levels of detection required for indoor air pollution, medical diagnoses, or agricultural/industrial monitoring. Further, we have demonstrated that our sensors can be operated unattended for several days and provide reliable habitability data QEPAS: Demonstrated very compact robust sensors that can be used in industy-process streams, petrochemical and/or biomedical fields.
Task Progress:	Developed both technologies to the point where a select outside panel made a downselect and chose QEPAS for further development for deployment on ISS as a technology demonstration. QEPAS sensor and electronic is ~ 0.8 l total volume. Demonstrated ~ 60 ppb for formaldehyde but could go even lower. The downselection was made after an extensive system engineering study at JPL, using Team X resources to develop strawman instrument designs.
Bibliography Type:	Description: (Last Updated: 11/02/2007)
Articles in Peer-reviewed Journals	Kosterev AA, Tittel FK, ,Serebryakov DV, Malinovsky AL, Morozov IV. "Applications of quartz tuning forks in spectroscopic gas sensing." Review of Scientific Instruments 2005 Apr;76(4):043105. http://dx.doi.org/10.1063/1.1884196 , Apr-2005
Articles in Peer-reviewed Journals	So SG, Wysocki G, Frantz JP, Tittel FK. "Development of digital signal processor controlled quantum cascade laser based trace gas sensor technology." IEEE Sensors Journal 2006 Oct;6(5):1057-67. http://dx.doi.org/10.1109/JSEN.2006.881350 , Oct-2006
Articles in Peer-reviewed Journals	Wysocki G, Kosterev AA, Tittel FK. "Influence of molecular relaxation dynamics on quartz-enhanced photoacoustic detection of CO2 at lambda =2 µm." Applied Physics B: Lasers and Optics 2006 Nov;85(2-3):301-6. http://dx.doi.org/10.1007/s00340-006-2369-9 , Nov-2006
Articles in Peer-reviewed Journals	Lewicki R, Wysocki G, Kosterev AA, Tittel FK. "Carbon dioxide and ammonia detection using 2 µm diode laser based quartz-enhanced photoacoustic spectroscopy." Applied Physics B: Lasers and Optics 2007 Mar;87(1);157-62. http://dx.doi.org/10.1007/s00340-006-2474-9 , Mar-2007